

Advanced Databases

7

Data Warehouses

An Environment for Data Analysis

- **DATA WAREHOUSE**

- A structured description of all those data that are necessary for a strategic analysis of the trends and the behaviour of a firm

- **ON-LINE ANALYTICAL PROCESSING (OLAP)**

- The name given to analysis activities (it is contrasted to On Line Transaction Processing, OLTP)

Data analysis

OLTP vs OLAP

The problem

- A promise of relational technology “flexible data access”:
 - A tool for the end user
 - to express all types of query
- Relational technology did not provide:
 - Complexity and rigidity of applications
 - Emphasis to OLTP
- Consequences:
 - Data volumes for operational procedures
 - Limited use of data for strategy

Functional separation

- Operations environments:
«on line» data management, to support update transactions

On Line Transaction Processing (OLTP)

- Analysis environment:
«off line» data management, dedicated to statistical queries and analyses

On Line Analytical Processing (OLAP)

OLTP: On Line Transaction Processing

- Traditional transaction processing, which automatizes the operational procedures of the organization
 - Predefined and relatively simple operations
 - Each operation involves «a few» data items
 - Detailed data, up to date
 - «Acid» transactional properties (atomicity, consistency, isolation, durability) are essential

OLAP: On Line Analytical Processing

- Data processing for decision support
 - Complex and random (unpredictable) operations
 - Each operation can involve multiple data
 - Aggregated, historical, and potentially not up-to-date data
 - «Acid» properties are not relevant, because operations are read-only

OLTP vs OLAP

- Configuration of a system dedicated to one of them is manageable
- It is extremely difficult to manage both workloads on the same system
- A variety of reasons:
 - Dishomogeneous users and requirements
 - Technical aspects

OLTP e OLAP

	OLTP	OLAP
User	clerk	manager
Function	Daily operations	Decision support
Design	Application oriented	Data oriented
Data	current, updated, detailed, relational, homogeneous	historical, aggregate, multidimensional, heterogeneous
Use	repetitive	casual
Access	read-write, indexed	read, sequential
Unit of work	Short transaction	Complex query
#Acc. records	tens	millions
#users	>thousand	tens/hundreds
Size	100MB – 1GB	>1TB
Metrics	tps	qph/response time

Technical reasons of the OLTP/OLAP conflict

- Lock conflict
 - OLTP: many fast transactions with exclusive locks
 - OLAP: a few long transactions with shared locks
 - OLTP+OLAP
 - either the OLTP transactions are delayed
 - or OLAP queries cannot be executed

Technical reasons of the OLTP/OLAP conflict

- Index use
 - OLTP: a few and only when clearly needed
 - OLAP: many, to deal with any possible use
 - OLTP+OLAP
 - Either OLTP transactions are slowed down by the need to update many indexes
 - or OLAP queries do not have available the indexes they need

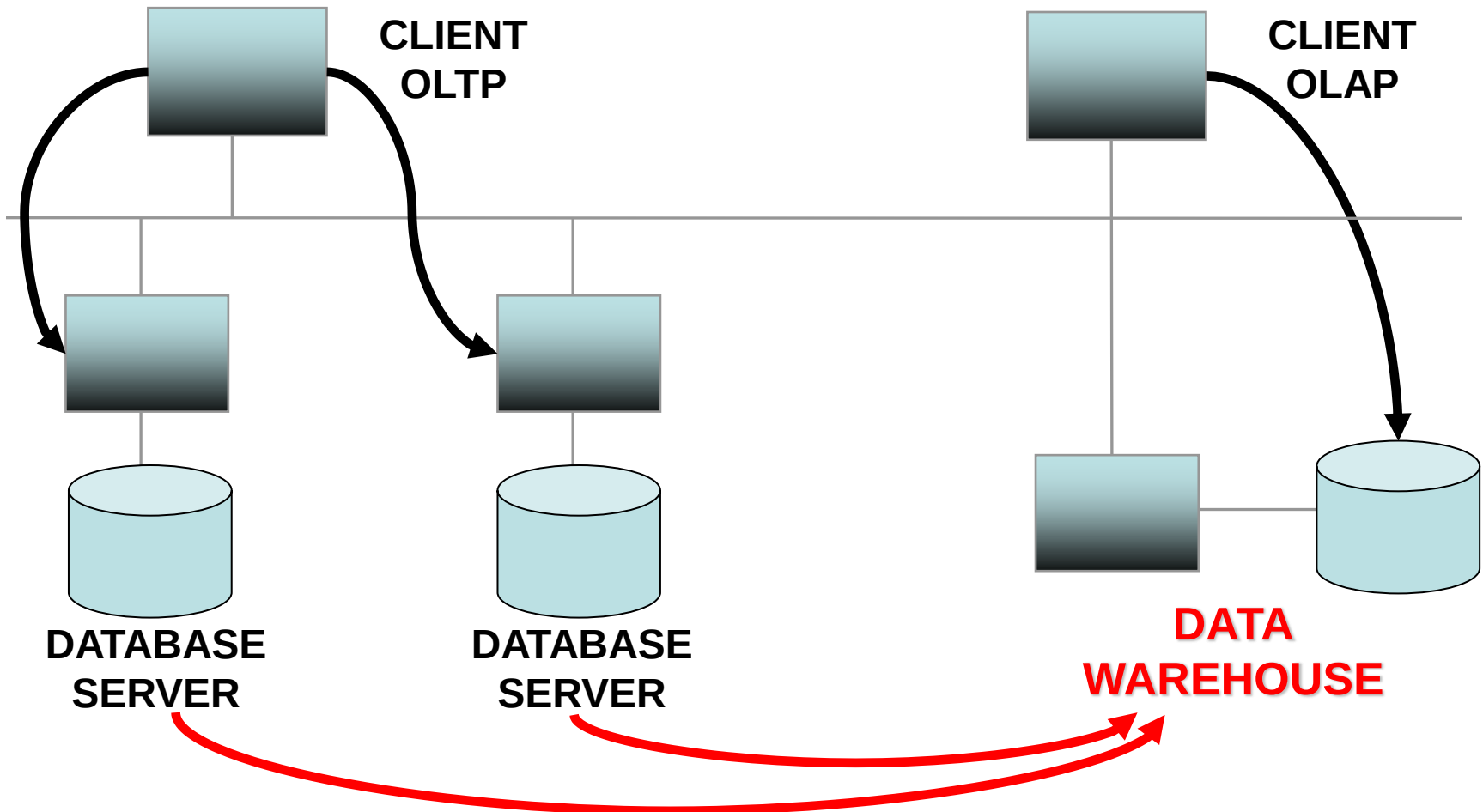
Technical reasons of the OLTP/OLAP conflict

- Query precomputations (i.e., materialized views)
 - OLTP: rarely, for consistency and load requirements
 - OLAP: key aspect to improve query response times
- Differences in the logical model
 - OLTP: high fragmentation and a large number of normalized tables
 - OLAP: few denormalized tables
- Different join algorithms

Considerations on the OLTP/OLAP conflict

- The conflict is intrinsic
- It does not disappear with the increase in computational power, rather it can increase
- The best solution consists in the separation of the two environments
 - This leads to the creation of Data Warehouses
- Asynchronicity of updates is fundamental

Interaction between OLTP and OLAP



Data Warehouse

Analysis environment: Data Warehouse

- **DATA WAREHOUSE:**
organized description of all the data required for a strategic analysis of the organization behavior
- **Techniques:** multidimensional analysis,
data mining

Data warehouse

- A data base
 - Mainly used to support management decisions
 - integrated — organization-level, not department-level
 - data-oriented — not application-oriented
 - historical — with a wide temporal coverage and (usually) attributes on the time dimension
 - not volatile — data are loaded and updated off-line
 - separate from operational data bases

... integrated ...

- The data of interest originate from all possible data sources — each data item can come from one or more of them
- The data warehouse represents data in a univocal way — reconciling the heterogeneity of the distinct representations
 - names
 - codes
 - multiple representations

... data-oriented ...

- Operational data bases are built to support specific operational processes and applications
 - production
 - sales
- The data warehouse is built around the major entities of the organization information resources
 - product
 - client

... historical data ...

- Operational data bases keep the current value of information
- Temporal horizon is in the order of a few months
- In data warehouses it is of interest to analyze the historical evolution of the data
- Temporal horizon is in the order of years

... not volatile ...

- In an operational data base, data are
 - Accessed, inserted, modified, deleted
- a few records at a time
- In a data warehouse, we have
 - «daily» access operations and queries
 - «nightly» load and update operations
- which are applied over millions of records

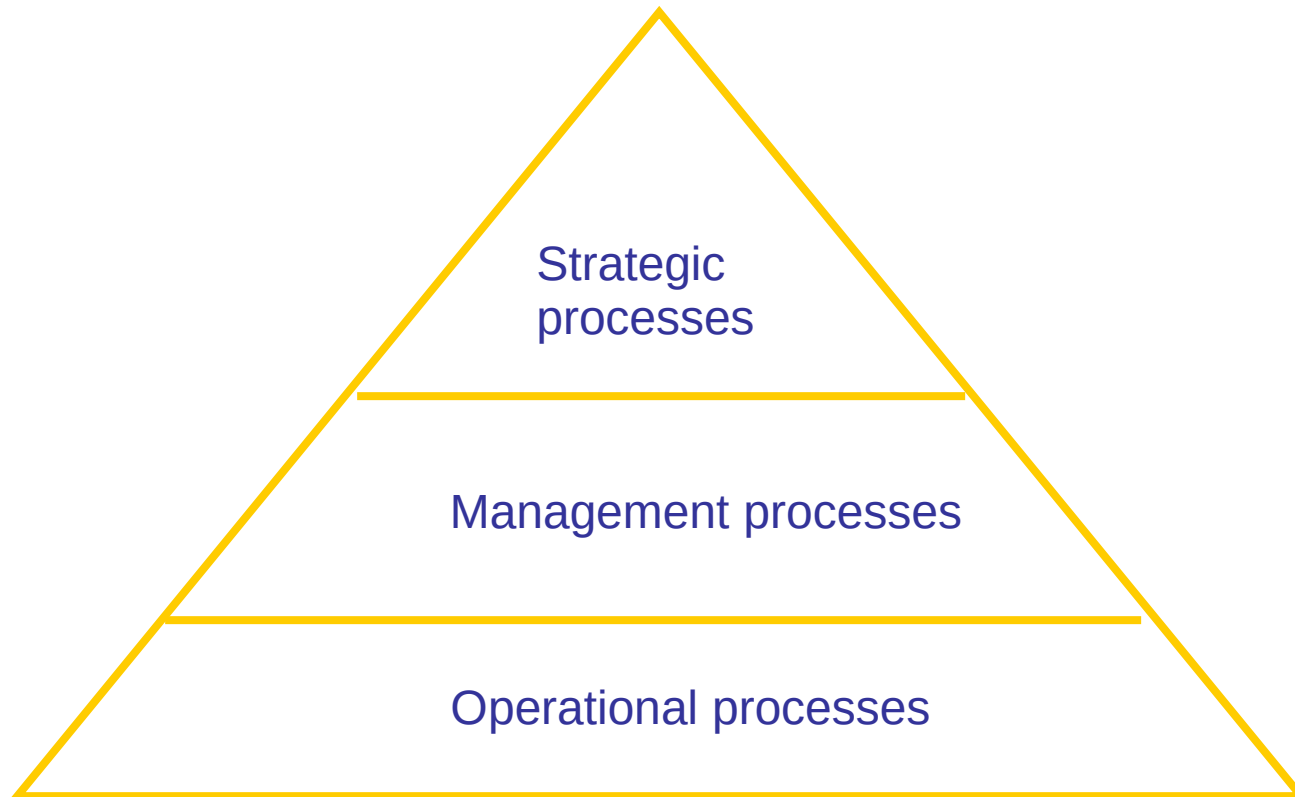
... a separate data base ...

- Due to several reasons
 - technical reasons we saw
 - There is no single operational database with all the data needed for analysis
 - The data base must be integrated
 - The data are in any case different
 - Historical data must be maintained
 - Aggregate data must be stored
 - Data analysis requires special organizations and access methods
 - General negative performance impact if there is no separation

Critical success factors

- Data replication must not have a great impact on the transactional system
- Data must be loaded in the assigned time window
- The solution must be scalable
- It must be accepted by users (presentation)
- Replicated data must be correct
- Standards should be used
- The data model should be consistent with reality

Processes

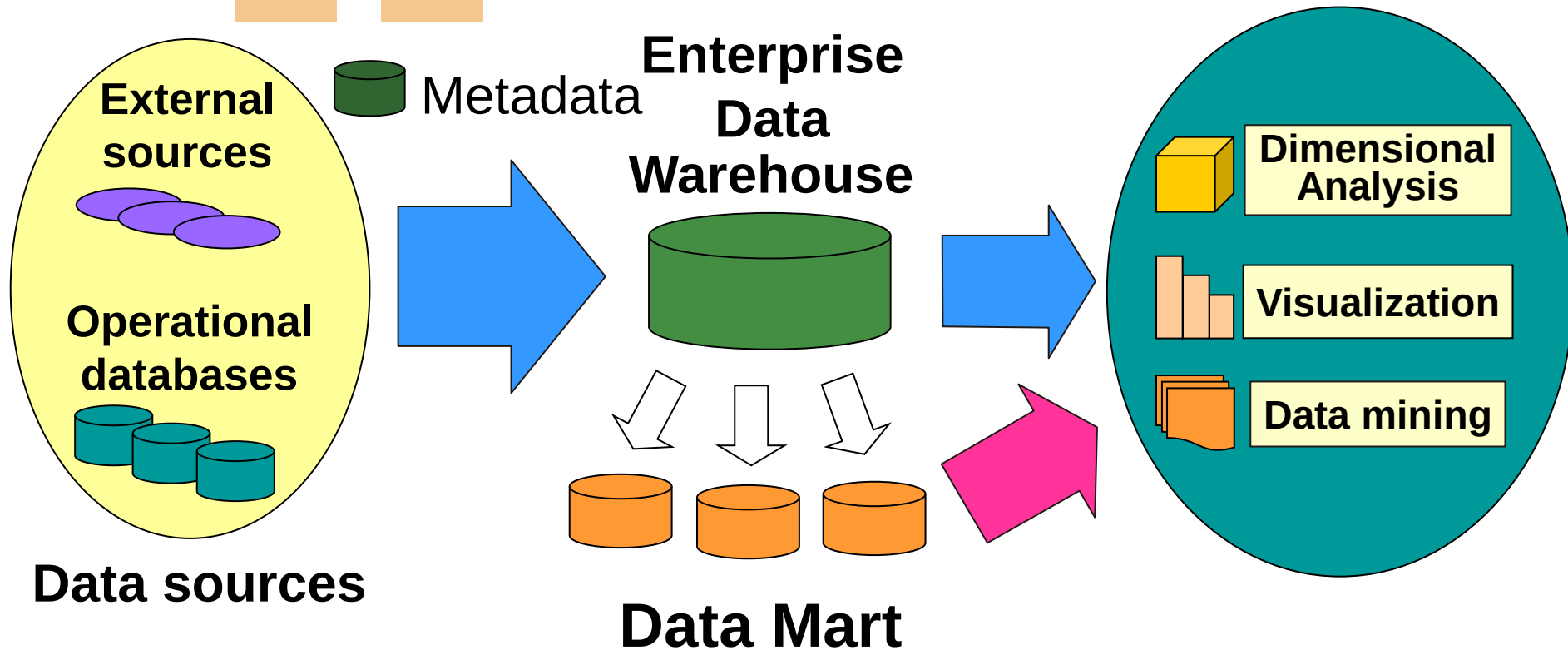


Management control systems

- MCSs share some features with OLAP applications
 - Mainly read-only
 - Need to compute aggregates
 - Comparison with historical series
- But they are different!
 - Static queries (not ad-hoc queries, rather fixed reports)
 - Need to keep the detail of the schema
- They can share information, but they must be distinct projects

An Architecture for Data Warehousing

Monitoring & Administration



DW and Data Marts

- A DW often integrated several Data Marts
- Users typically access a specific Data Mart
- Data Marts share data among themselves
- Each Data Mart is responsible of a specific aspect of the organization reality

Multidimensionale Model

Star model

- The *star model* is used for each Data Mart
 - Also known as *multi-dimensional schema*
- It is a conceptual model which introduces some restrictions
- Advantages:
 - Availability of suitable specific query interfaces
 - Good performance
 - Straightforward design of the relational schema

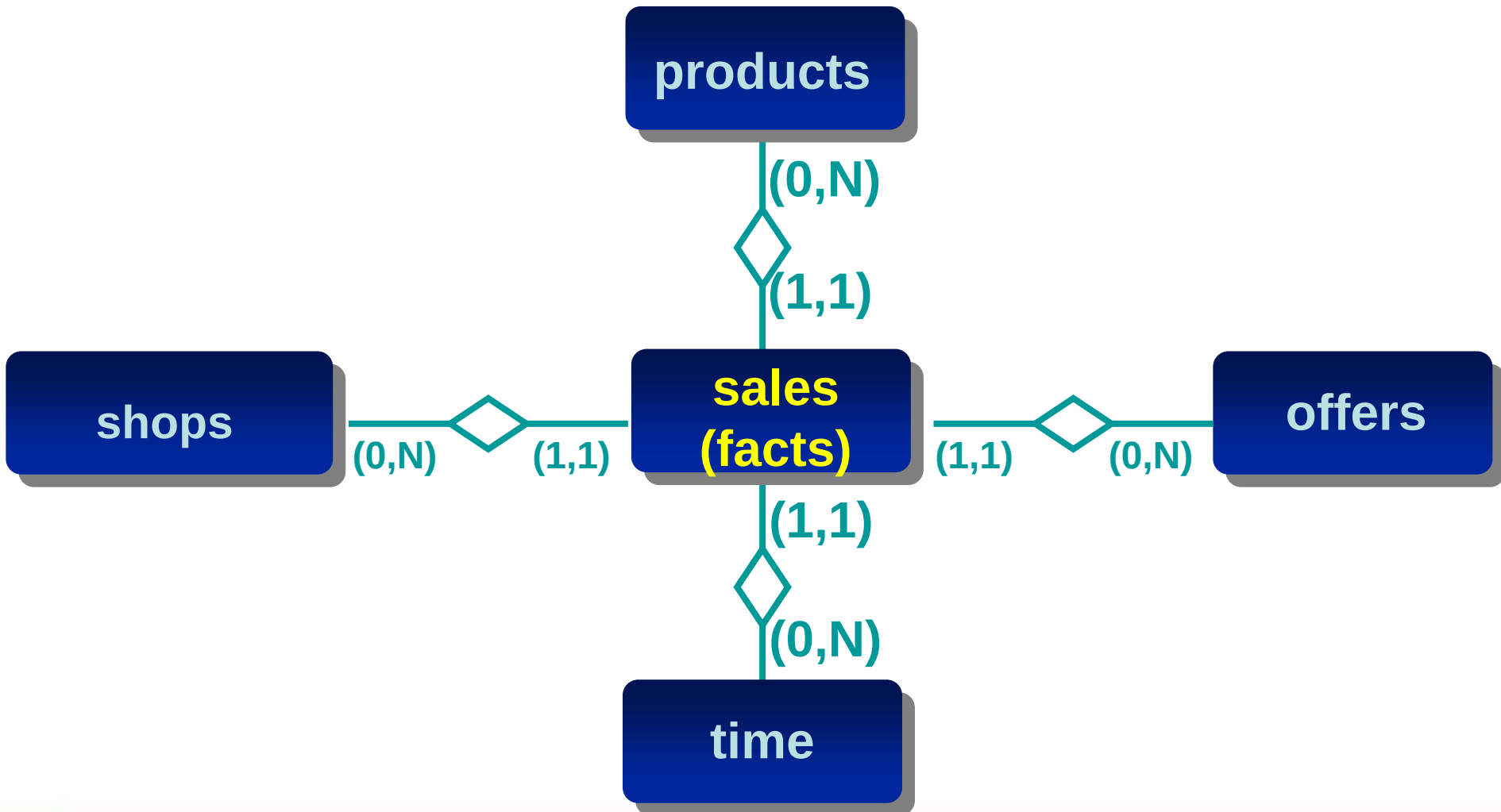
Multi-dimensional representation

- Relevant concepts:
 - **fact** — an aspect which is crucial for the analysis
 - **measure** — an atomic property of a fact
 - **dimension** — a specific perspective for the analysis

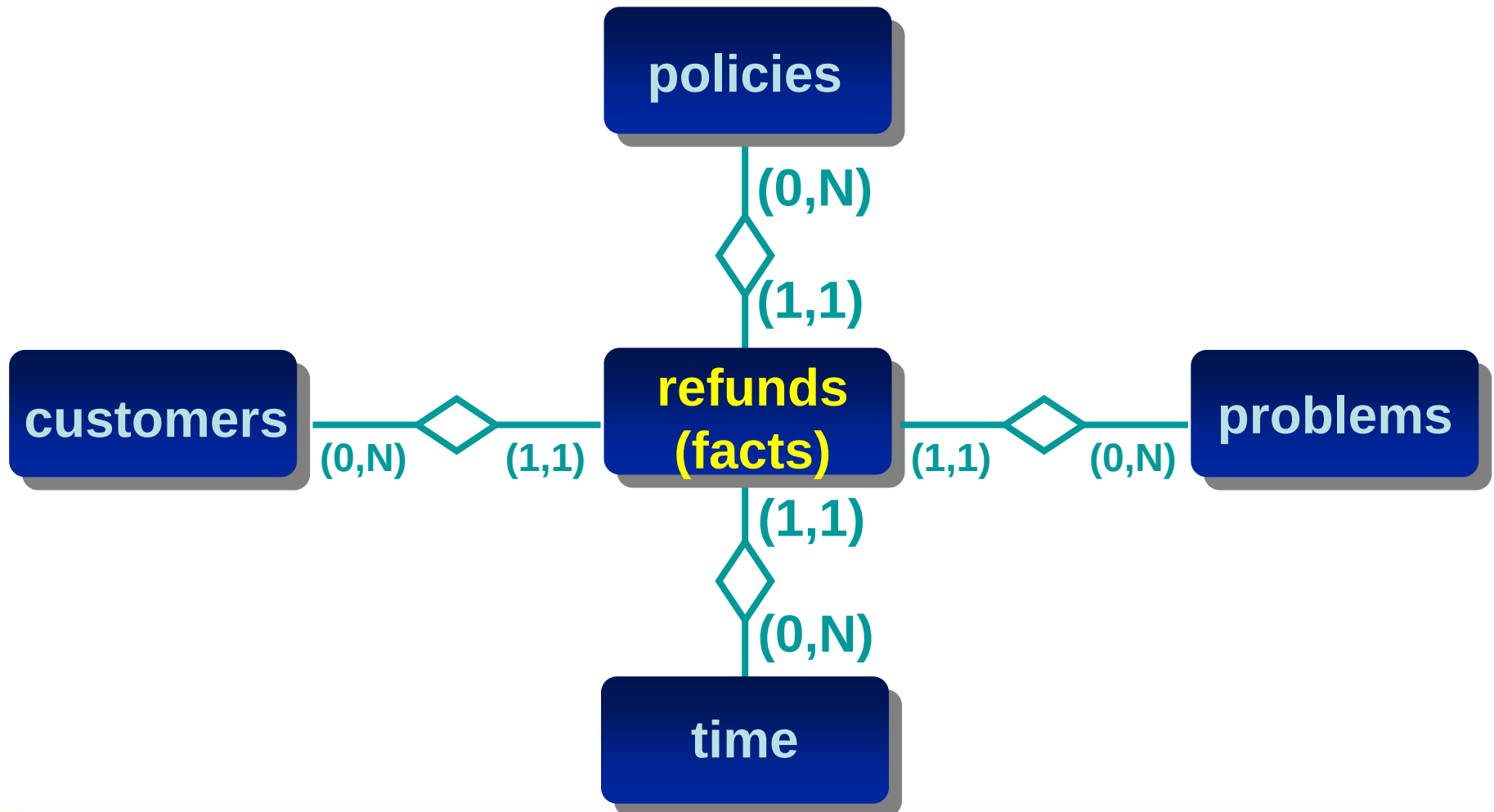
Examples of facts/measures/dimensions

- Retail shops:
 - Sales
 - Quantity, price
 - Product, time, zone
- Telephone service:
 - Phone call
 - Cost, duration
 - Caller, answerer, time

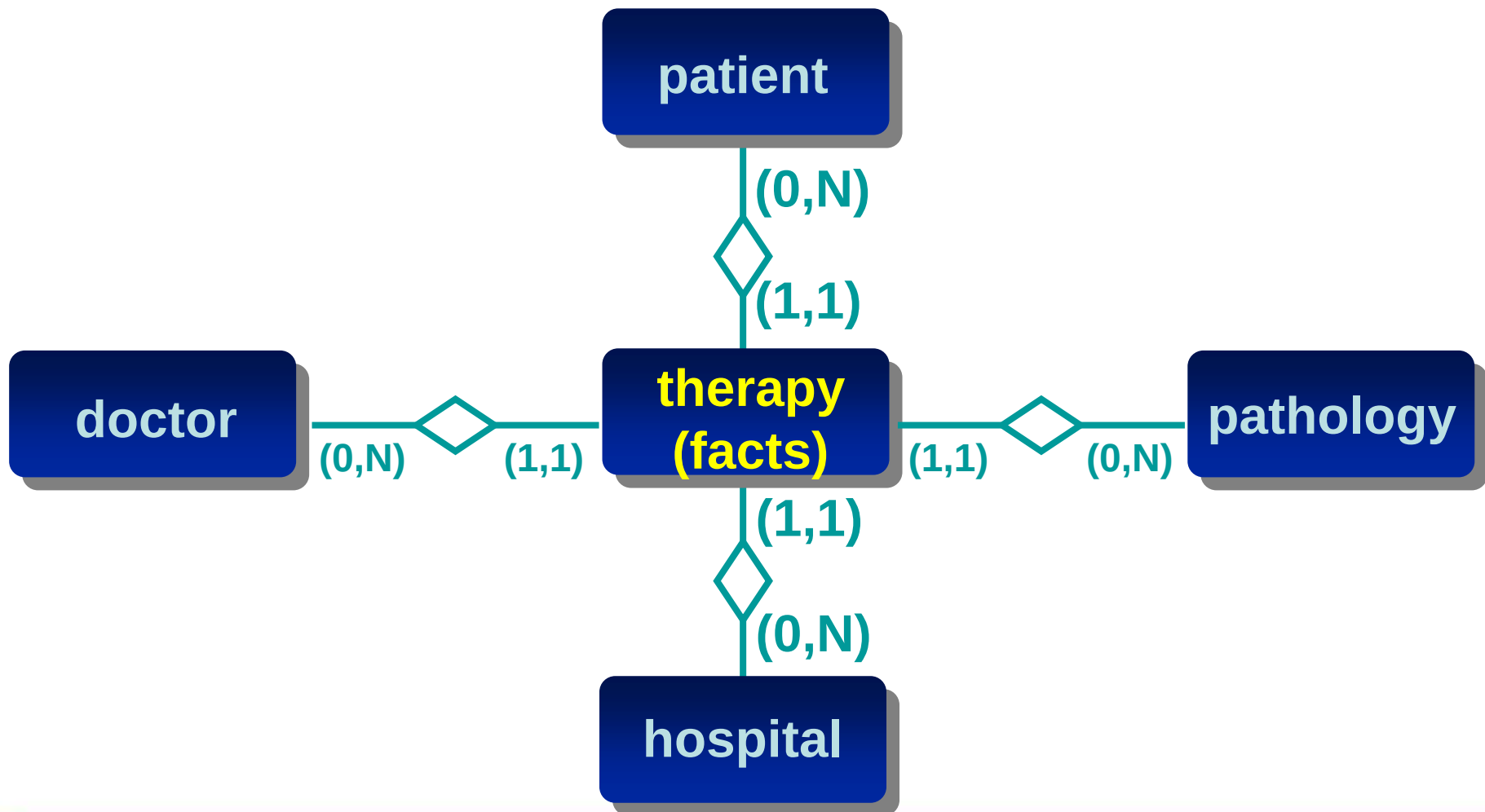
An example: sales management



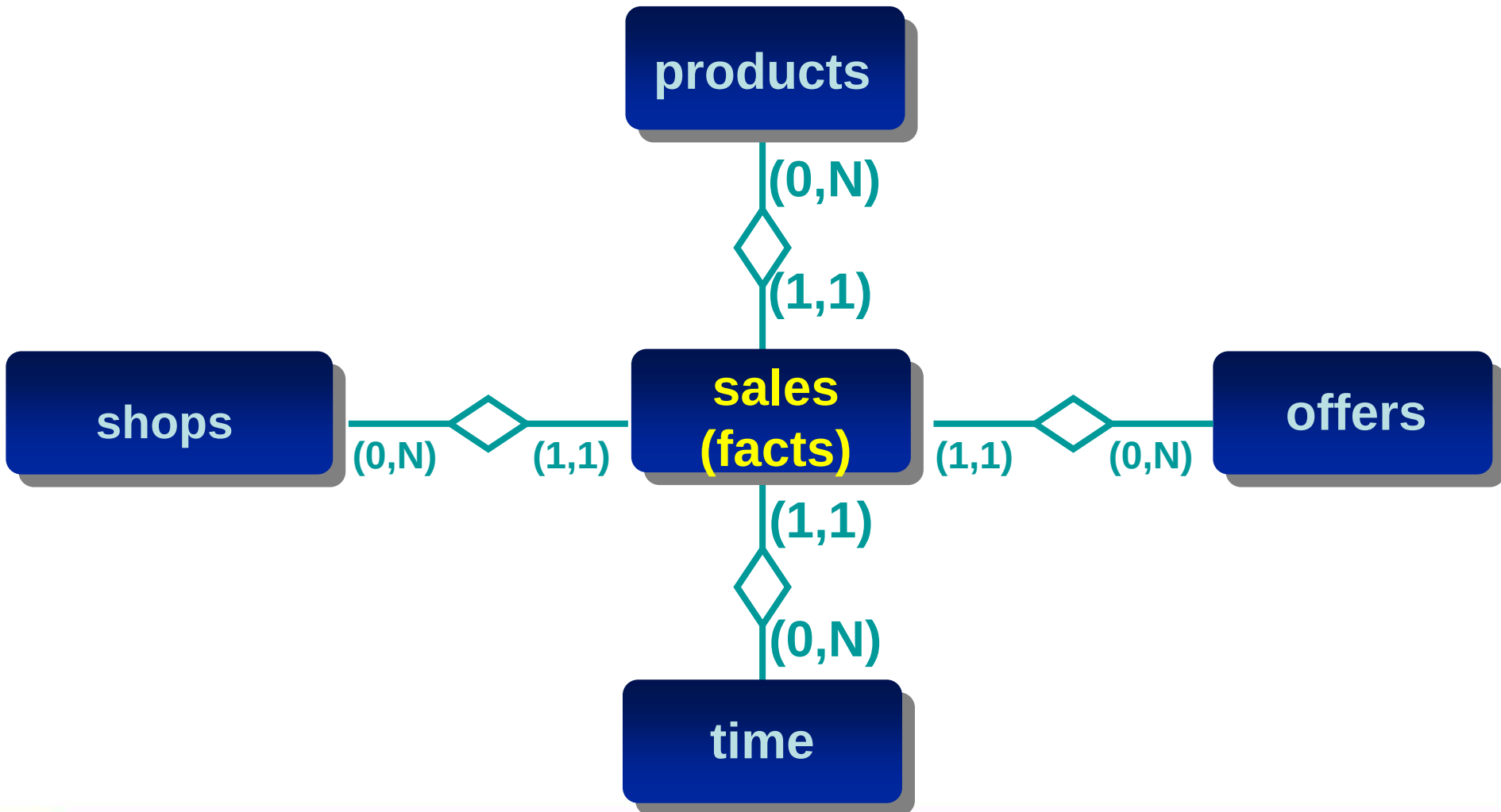
Another example: reimbursements



Another example: therapies



Consider again the sales management schema



Facts: sales**Product-ID****Shop-ID****Time-ID****Offer-ID****Total-proceeds****Quantity****Unit-proceeds**

First dimension: products

Product-ID

Category

Sub-Category

Brand

Packing

Weight

Size

Provider

Second dimension: shops

Shop-ID

Name

Address

City

Sales-District

Phone

Manager-Name

Size

Logistics

Third dimension: time

Time-ID

Day-in-Week

Day-in-Month

Day-in-Year

Week-in-Month

Week-in-Year

Month-in-Year

Season

Flag-WorkingDay

Flag-Sunday

Fourth dimension: offers

Offer-ID

Offer-name

Discount-Type

Discount-Percentage

Advertisement

Flag-Coupon

Start-Date

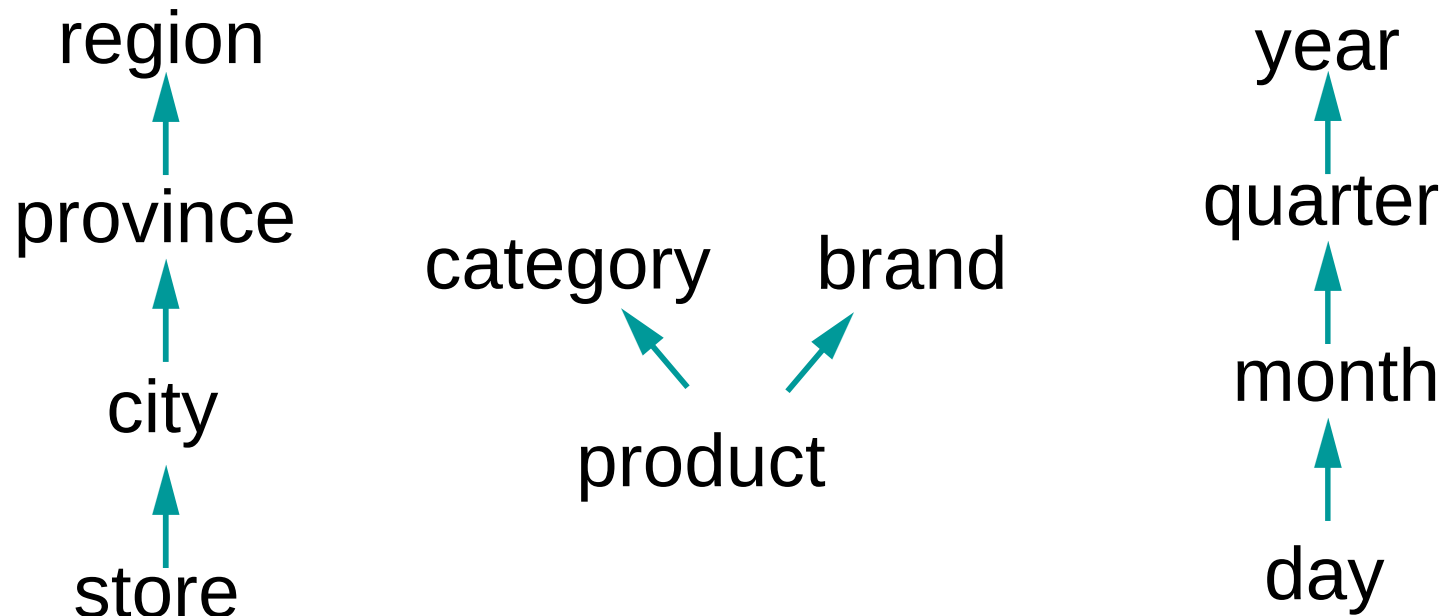
End-Date

Cost

Agency

Dimensions and hierarchies

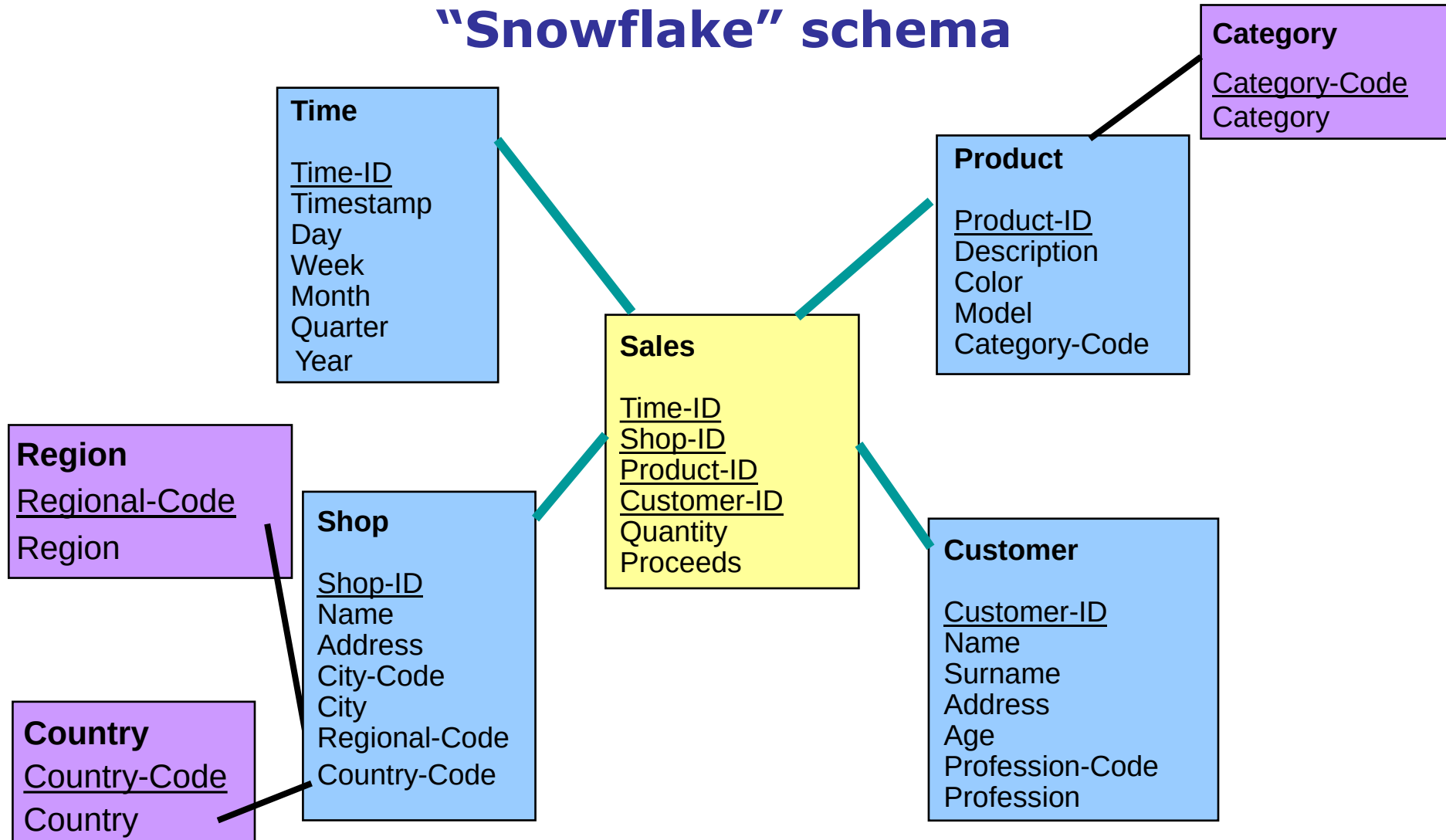
- Each dimension is organized in a hierarchy, which represent possible aggregation levels



Snow-flake model

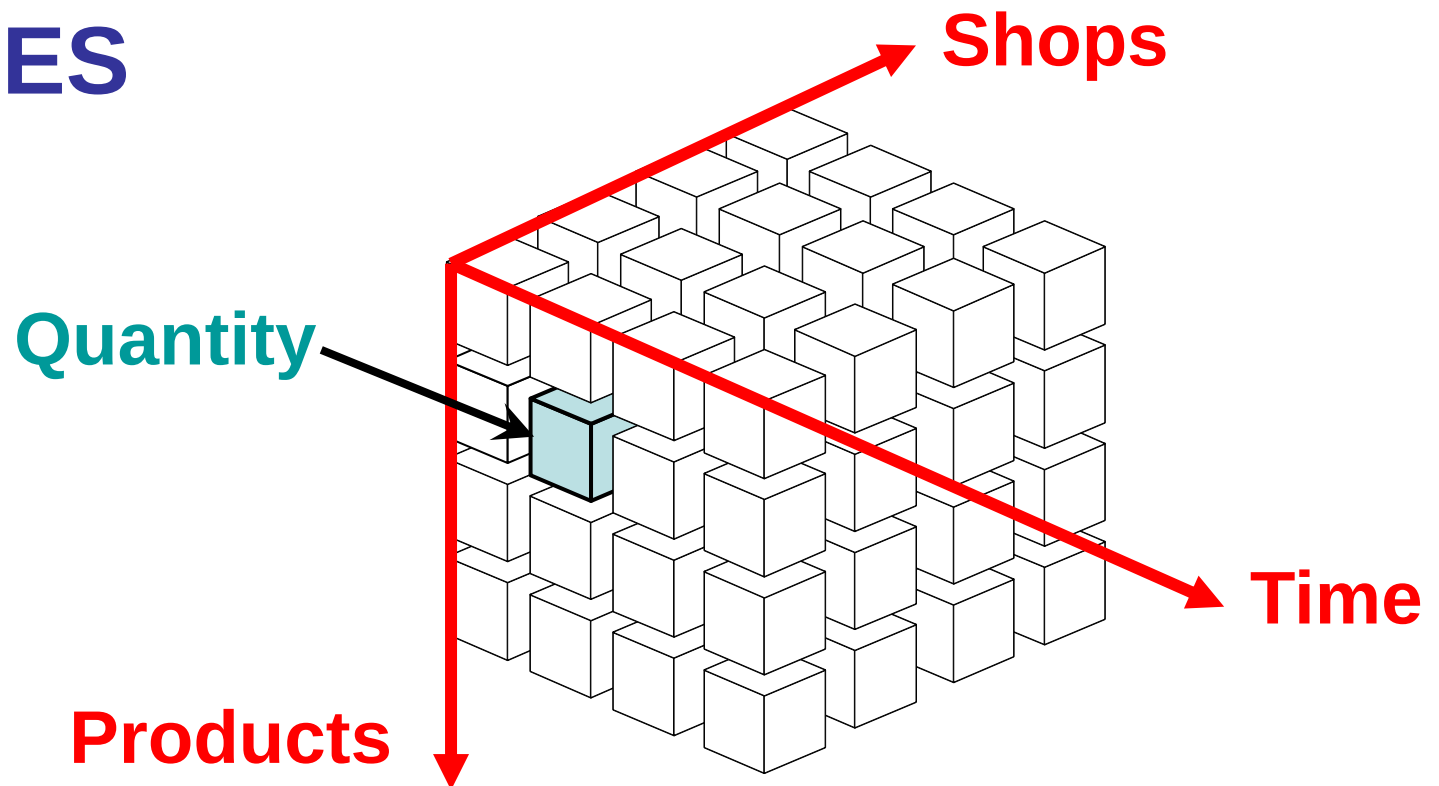
- It extends the star model
- It permits to reduce excessive redundancy in dimensions
- Starting from the fact table, it is always possible to reach all the dimension (sub)-tables, always passing through n:1 relationships

"Snowflake" schema

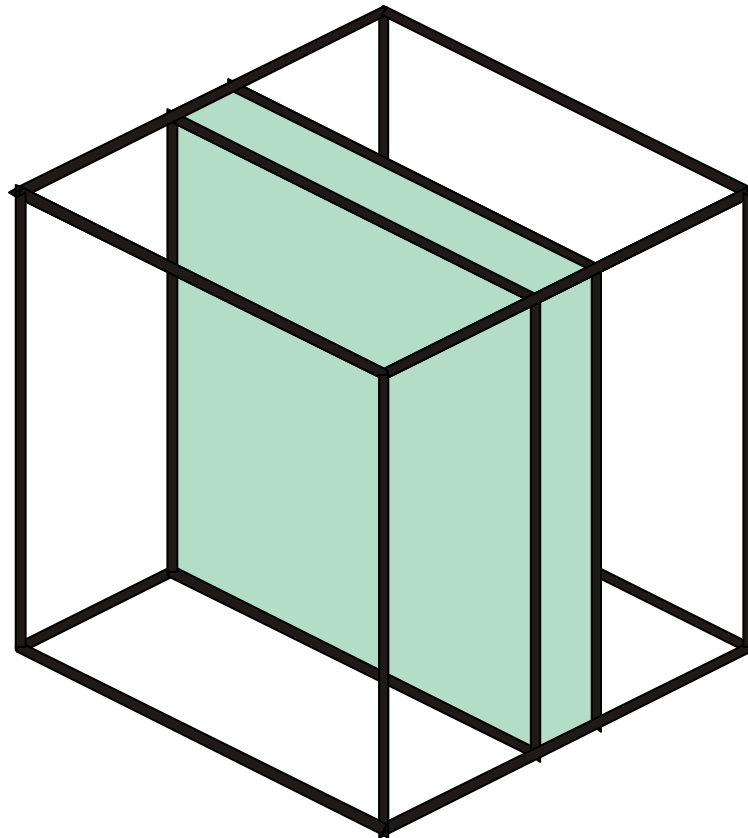


Multi-dimensional data representation

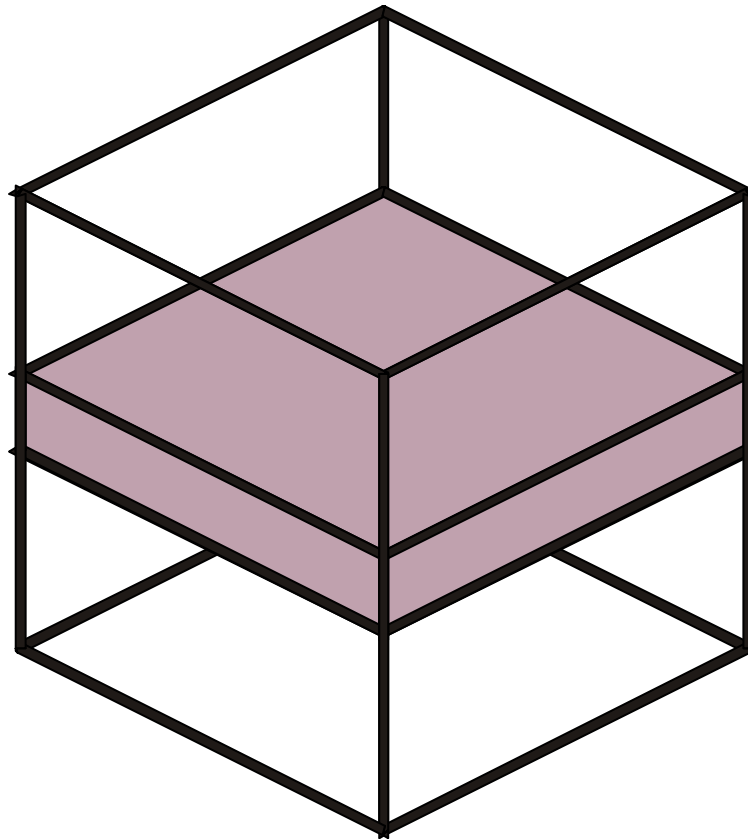
SALES



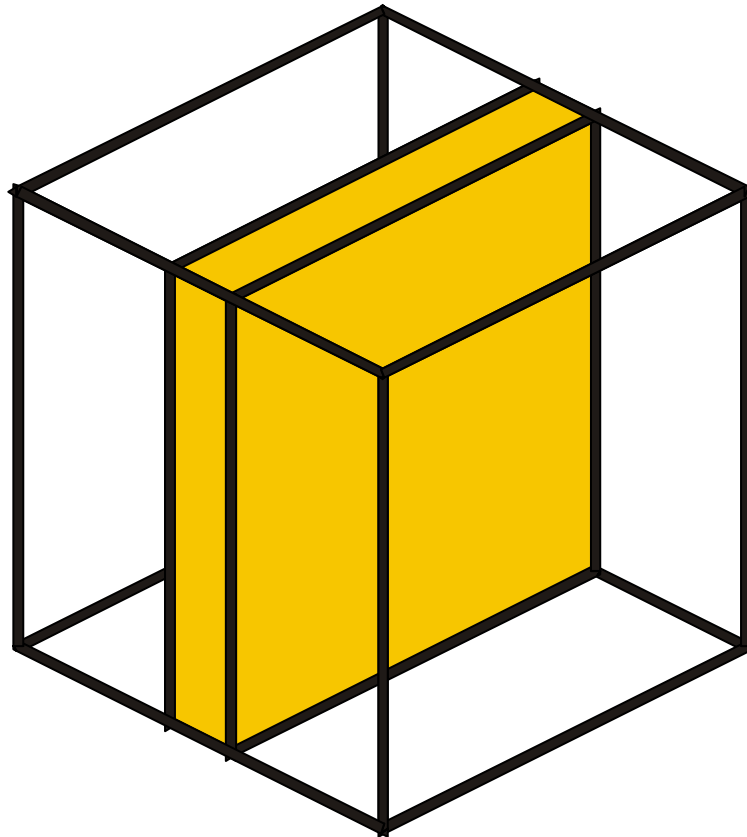
- The regional manager studies the sales of all products in all periods w.r.t. the shops of his region



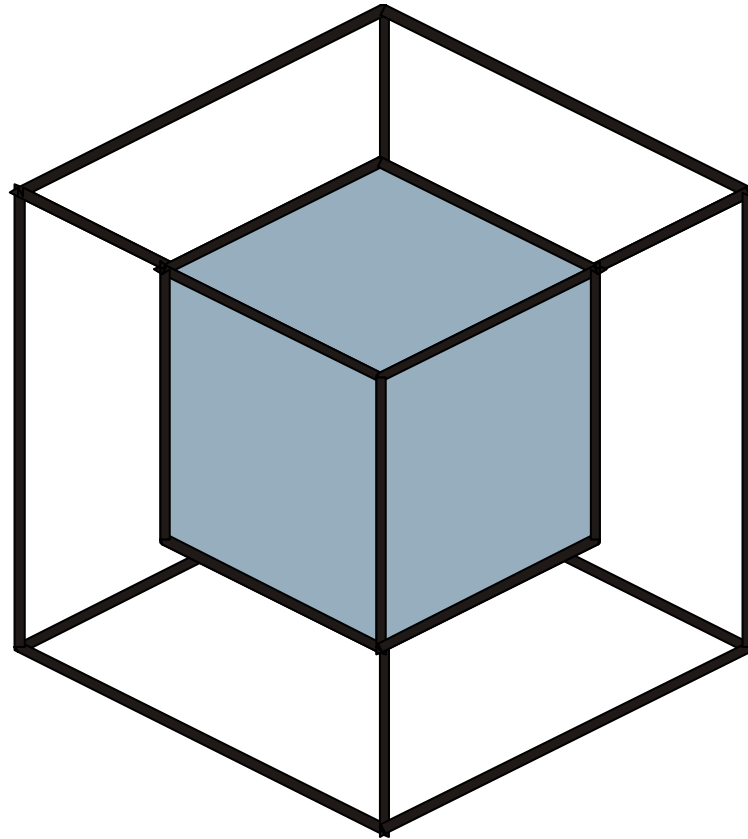
- The product manager studies the sales of a product in all periods and in all shops



- The financial manager studies the sales of all products in all shops, comparing the current period with the previous one



- The strategic manager focuses on one category of products, one area and a limited period of time



Data Visualization

- Data are visualized and rendered graphically, so as to be easy to understand
- Common means of visualization:
 - Tables
 - Pie/doughnut charts
 - Column bar/histograms
 - Line charts
 - 3D surfaces
 - Bubble charts
 - Area blocks
 - Cylinders/cones/pyramids
 - ...

Example of query with a browser

Offer	period	zone	product	dimension of analysis
Pay 2 & buy 3	March	north	milk	total-proceeds
40% discount	April	east	bread	quantity
20% discount	May	west	pasta	unit-proceeds
1-free-mug (...)				
.....				
	February/ April		pasta	sum(proceeds) sum(quantity)

The “same” query in SQL

```
select  c1, c2, aggr(c3), aggr(c4)
from    facts, dim1, dim2
where   join-predicate(facts, dim1)
        and join-predicate(facts, dim2)
        and selection-predicate(dim1)
        and selection-predicate(dim2)
group by c1, c2
order by c1, c2
```

Result

Month	product	Sum of proceeds	Sum of quantity
February	pasta	110.000	45.000
March	pasta	95.000	50.000
April	pasta	105.000	51.000

Operations over multi-dimensional data

- **Roll up** — aggregates data
 - Sums up the sale quantity over last year per each region and product category
- **Drill down** — disaggregates data
 - For one particular product category in a region, “unrolls” and shows in detail the sale quantities of each day in each shop
- **Slice & dice** — selection and projection
- **Pivot** — change the orientation of the data cube

Drill Down: adding one dimension (Zone)

Month	Product	Zone	Sum of quantity
February	pasta	north	15.000
February	pasta	east	17.000
February	pasta	west	13.000
March	pasta	north	18.000
March	pasta	east	18.000
March	pasta	west	14.000
April	pasta	north	18.000
April	pasta	east	17.000
April	pasta	west	16.000

Roll-up: removing one dimension (Month)

Product	Zone	Sum of quantity
pasta	north	51.000
pasta	east	52.000
pasta	west	43.000

Aggregate queries

- **Examples:**
 - Total proceeds for each product category in each shop in each day
 - Total monthly proceeds in each shop
 - Total monthly proceeds for each product category in each shop
 - Average monthly proceeds for each category (calculated over all shops)

Aggregates in SQL: data cube

- Expresses all possible aggregations of the tuples of a table
- Uses a new purpose-specific *polymorphic* value: **ALL**

Data cube in SQL

```
select Model, Year,  
        Color, sum( Quantity )  
from Sales  
where Model in ('Fiat','Ford')  
       and Color = 'Red'  
       and Year between 1994 and 1995  
group by Model, Year, Color  
with cube
```

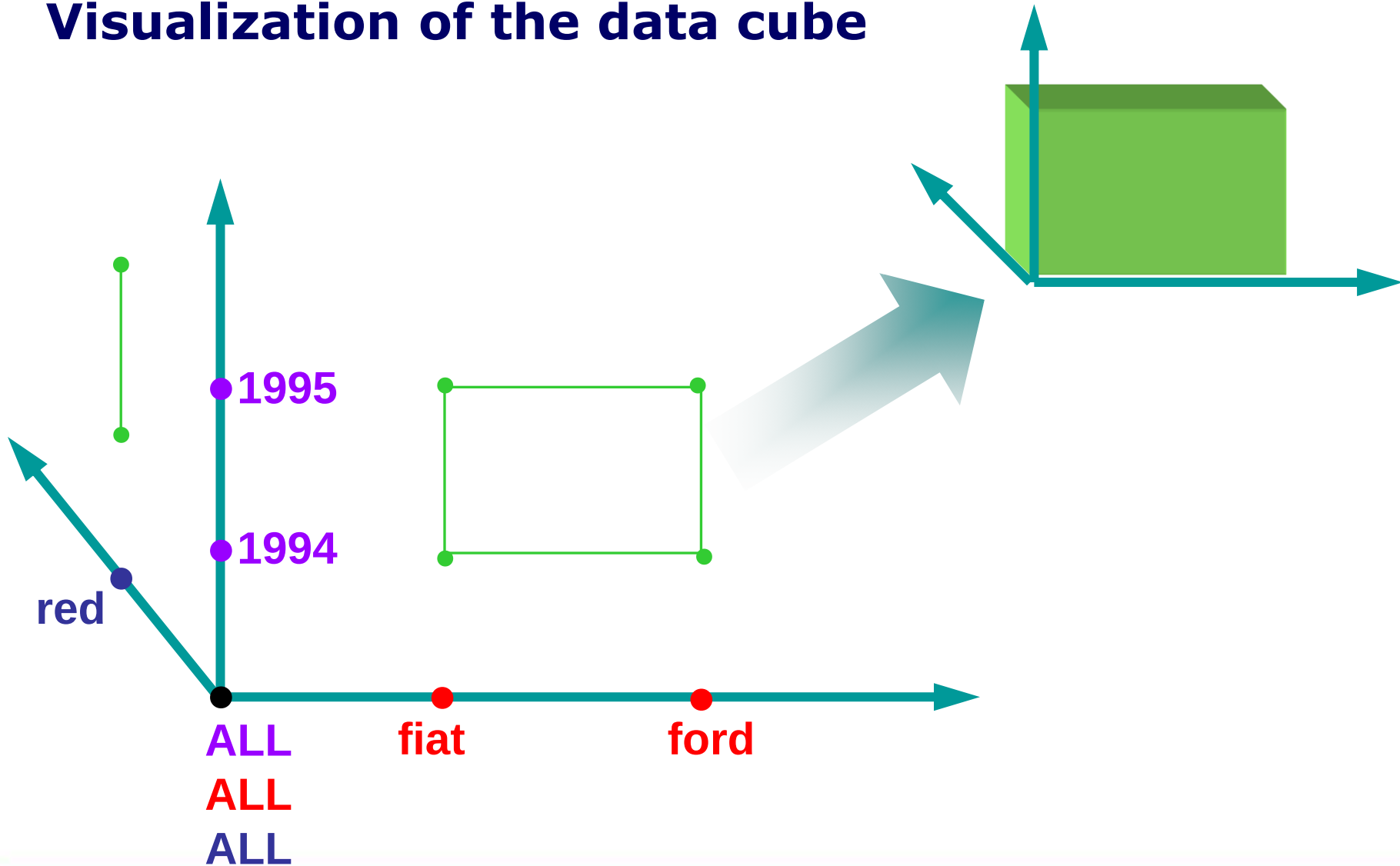
Relevant facts

Model	Year	Color	Quantity
fiat	1994	red	50
fiat	1995	red	85
ford	1994	red	80

Data cube results:

model	year	color	sum (quantity)
fiat	1994	red	50
fiat	1995	red	85
fiat	1994	ALL	50
fiat	1995	ALL	85
fiat	ALL	red	135
fiat	ALL	ALL	135
ford	1994	red	80
ford	1994	ALL	80
ford	ALL	red	80
ford	ALL	ALL	80
ALL	1994	red	130
ALL	1995	red	85
ALL	ALL	red	215
ALL	1994	ALL	130
ALL	1995	ALL	85
ALL	ALL	ALL	215

Visualization of the data cube



Roll up in SQL

```
select Model, Year,  
        Color, sum( Quantity )  
from Sales  
where Model in ('Fiat', 'Ford')  
       and Color = 'Red'  
       and Year between 1994 and 1995  
group by Model, Year, Color  
with roll up
```

Roll up results:

Model	Year	Color	sum(Quantity)
fiat	1994	red	50
fiat	1995	red	85
ford	1994	red	80
fiat	1994	ALL	50
fiat	1995	ALL	85
ford	1994	ALL	80
fiat	ALL	ALL	135
ford	ALL	ALL	80
ALL	ALL	ALL	215

Typical Size of a Data Warehouse

time: 730 days

shops: 300

products: 30.000

daily sales (avg number of sales of a product in a daily time slot for each shop): 3.000

offers: at most one per product on sale

sales: $730 * 300 * 3000 * 1 = 657$ millions

**Size fact table: $657 \text{ millions} * 8 \text{ attributes} * 4 \text{ Byte}$
= **21 GB****

Data warehouse design

Data Warehouse design

- The design of a data warehouse differs from the design of an operational database
 - Data have different features
 - Constrained by existing databases
 - Driven by different design criteria
- Emphasis on abstraction and conceptual clarity
 - Few entities
 - Wide coverage
- Main activities
 - analysis — of existing data sources
 - integration
 - design — conceptual, logical and physical

Integration of data sources

- Data source integration is the merging of data represented in multiple sources into a unique global data base that represents the global information assets of the organization
- Main goal of integration is the identification of all the portions of the distinct data sources that refer to the same aspect of the described reality, to unify its representation
- The approach is directed toward the identification, analysis and resolution of conflicts — terminological, structural, and codification

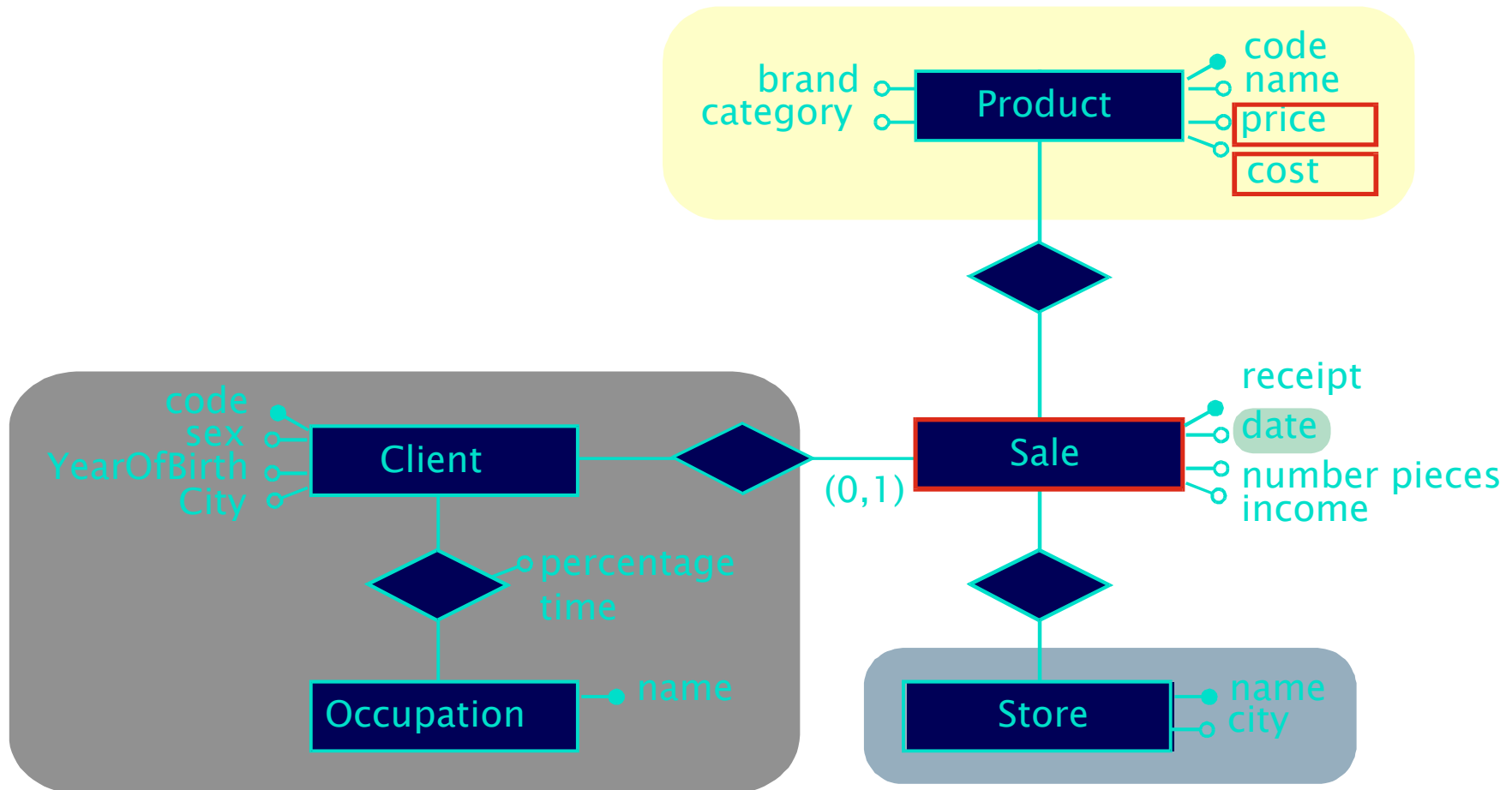
Examples of conflicts

- Conflicts associated with the codification of information
 - A «sex» attribute can be represented as:
 - With a single char — M/F
 - With a digit — 0/1
 - Implicit in the fiscal code
 - Not represented
 - First and last names of a person
 - "Mario", "Rossi"
 - "Mario Rossi"
 - "Rossi, Mario"
 - "Rossi, M."

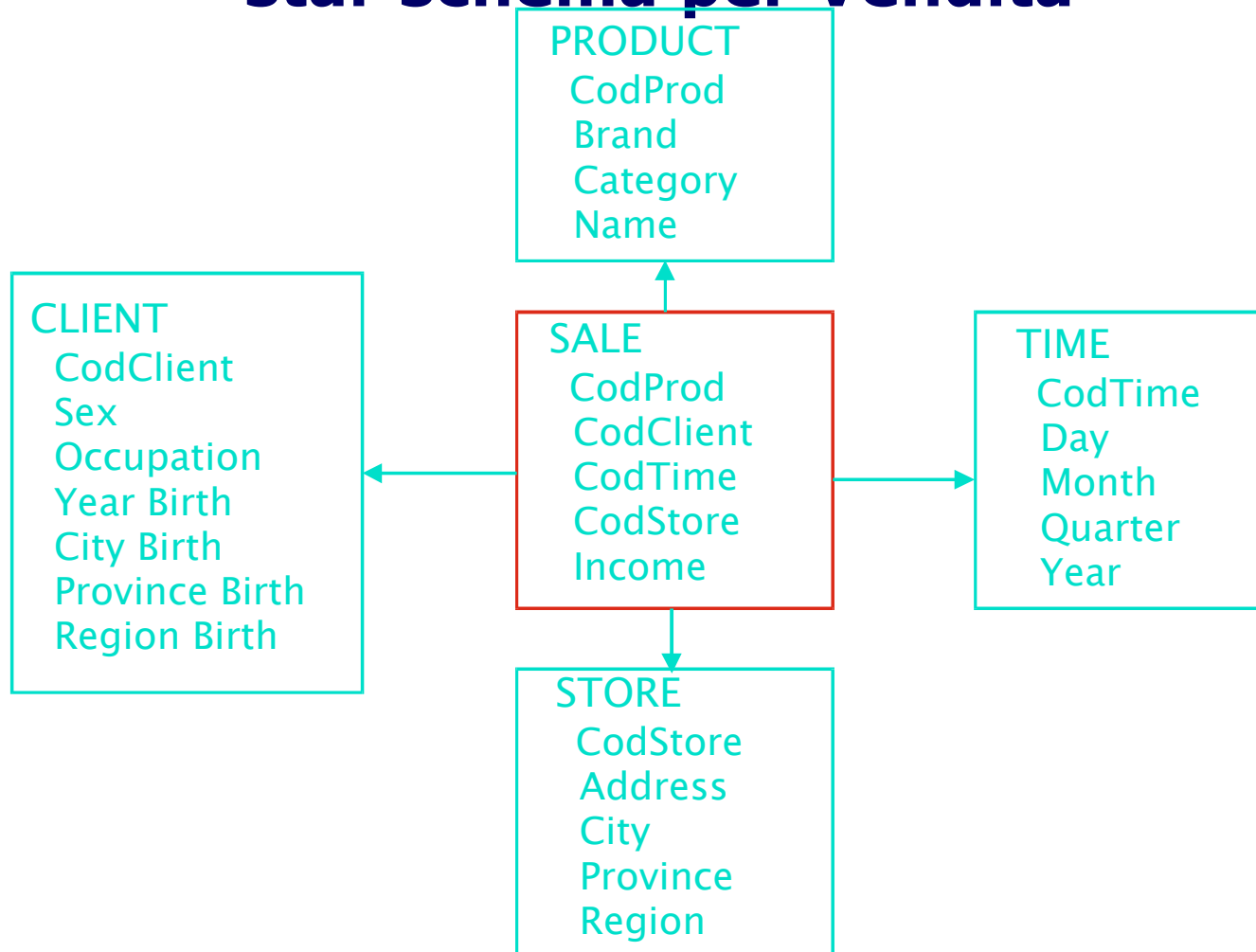
Identification of data marts

- Usually: several data marts present in the data warehouse
- Activities
 - Identification of facts, measures and dimensions
 - Restructuring of the conceptual schema
 - Representation of facts by way of entities
 - Identification of new dimensions
 - Refinement of the dimension levels
 - Derivation of a dimensional graph (star or snow-flake, depending on the scenario)
 - Logical and physical design of data marts, and of the population mechanism

Identification of facts and dimensions



Progettazione logica: star schema per Vendita



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Data Warehouses

Technical aspects

Classification of OLAP System

- **MOLAP** (**M**ulti-dimensional OLAP)
as alternative to
- **ROLAP** (**R**elational OLAP)
 - MOLAP: the internal data storage is not relational, so as to guarantee better **performance**
 - ROLAP: the relational storage guarantees the capability of managing large **volumes** of data

Specific OLAP Technologies

- **Bitmap Indexes**
 - Allow for efficient evaluation of OR and AND combinations of simple comparison predicates
- **Join Indexes**
 - Pre-computed joins between the table of facts and the tables representing the dimensions
- **Materialized views**
 - Those views are pre-computed, which can be used to answer most frequently asked queries

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Data Warehouses

Data Mining

Data mining

- Objective
 - Extract information *hidden* into data so as to support strategic decisions
- An inter-disciplinary task (and subject)
 - Statistics
 - Algorithmics
 - Neural networks
 - Fractals
 - ...

Applications of Data Mining

- Market analysis
 - Which products are purchased together or one before another? (basket analysis)
- Analysis of behaviours
 - Identify fraudulent credit card usage
- Forecasts
 - Foreseeing the cost of medical treatments
- Control
 - Industrial production errors

An example: sales analysis

Transaction	Date	Item	Qty	Price
1	12/17/95	ski-pants	1	140 €
1	12/17/95	ski-boots	1	180 €
2	12/18/95	T-shirt	1	25 €
2	12/18/95	jacket	1	300 €
2	12/18/95	ski-boots	1	70 €
3	12/18/95	jacket	1	300 €
4	12/19/95	T-shirt	3	25 €
4	12/19/95	jacket	1	300 €

Association Rules

- Association rules look for *regularity* within data
 - When a customer buys ski-boots, she also buys skis
- Structure of association rules:

Body* $\Rightarrow$ *Head

- *Body*: premise of the rule
- *Head*: consequence of the rule

An example of Association Rule

Diaper $\Rightarrow$ Beer

- 2% of all transactions contain both items
- 30% of transactions containing Diaper also contain Beer

Characteristics of Association Rules

- **Support**

- Probability that both Head and Body are in the same transaction [$P(H,B)$]

- **Confidence**

- Probability that the Head is in a transaction t , given that the Body **is** in t [$P(H|B)$, *conditional probability*]

- **Problem statement**

- Extract from a dataset all association rules with support and confidence over given thresholds

Examples of association rules

Body	Head	Support	Confidence
ski-pants	ski-boots	0.25	1
ski-boots	ski-pants	0.25	1
T-shirt	ski-boots	0.25	0.5
T-shirt	jacket	0.25	1
ski-boots	T-shirt	0.25	0.5
ski-boots	jacket	0.25	1
jacket	T-shirt	0.5	0.66
jacket	ski-boots	0.25	0.33
{ T-shirt, ski-boots }	jacket	0.25	1
{ T-shirt, jacket }	ski-boots	0.25	0.5
{ ski-boots, jacket }	T-shirt	0.25	1

Other Examples

- Items sold in the same special offer
- Items frequently purchased together in summer but not in winter
- Items frequently purchased together as in the shop they are arranged in a particular layout (adjacent, near, ...)
- Items purchased in consecutive transactions by the same customer

Sequential Patterns

- Input dataset:
 - All the transactions of a given customer
- Objective:
 - Find those sequences of items which are frequently contained into corresponding sequences of transactions, such that the frequency is over a given threshold

Examples

- “5% of customers bought a CD player in a transaction and some CDs in the following two transactions”
- “10% of the purchases of a television set is followed by the purchase of a video-recorder”
- Applications
 - Measure of the customer satisfaction
 - Special offers tailored for specific customer classes
 - Medicine (sequences of symptoms $\Rightarrow$ disease)

Discretization

- A continuous domain can be represented by means of a sequence of suitable intervals
 - Example: blood-pressure
 - High: >250
 - Medium: $>130, <250$
 - Low: <130
- *Objective*: find the correlation between the risk of infarction and blood-pressure with a given statistical significance
- *Advantages*:
 - Compact value representation
 - Determination of critical values
 - Facilitation of future data analysis

Classification

- Cataloguing a fact, concept, or phenomenon into a pre-defined class
- The phenomenon is described by elementary facts (atomic data, within **tuples**)
- The classifier is constructed and trained over a set of training data (**training set**)
- Classifiers are represented as **decision-trees**

Example: identify risky policies

POLICY (DrivingLicense, Age, CarType)

